

Road Accident Analysis

Data-driven insights for improving road safety in
Egypt

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Why Road Accident Analysis?

Road accidents represent a critical public health and economic challenge in Egypt, impacting thousands of lives annually.

Fatalities & Injuries: Permanent loss and long-term disability for individuals and families.

Economic Burden: Material losses and healthcare costs estimated in millions of dollars.



Project Objectives



Pattern Analysis Discover underlying trends and seasonal variations in accident data.



Risk Identification Pinpoint high-risk governorates and dangerous road segments.



Interactive Dashboards Enable decision-makers to explore real-time KPIs and metrics.



Strategic Recommendations Formulate data-backed interventions for road safety improvements.



Data Source & Overview

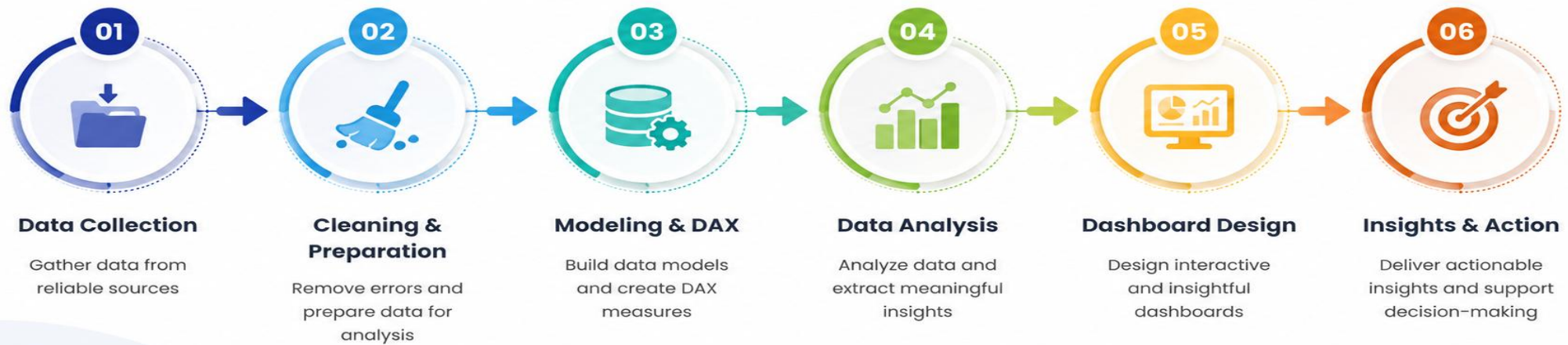


CAPMAS Dataset

Data provided by the Central Agency for Public Mobilization and Statistics (CAPMAS) for the year 2024.

- ✓ Accident & Casualty counts
- ✓ Governorate distribution
- ✓ Vehicle & Road User types
- ✓ Age group demographics

Project Workflow



Dashboard Overview

"The dashboard provides a comprehensive analysis where each component highlights a critical aspect of Egypt's road safety landscape."

Executive Overview

High-level KPIs including accidents, deaths, and injuries.

Victim Demographics

In-depth analysis of age groups, gender, and user types.

High Risk Roads

Spatial mapping of dangerous roads and hazard points.

Emergency Response

Evaluation of ambulance coverage vs. population density.



Egypt Road Traffic Accidents Dashboard

Executive Overview • 2024

This dashboard provides an executive summary of road traffic accidents in Egypt during 2024, highlighting severity, geographic distribution, and key trends

Cruelty Rate

6.83

Total injuries

76K

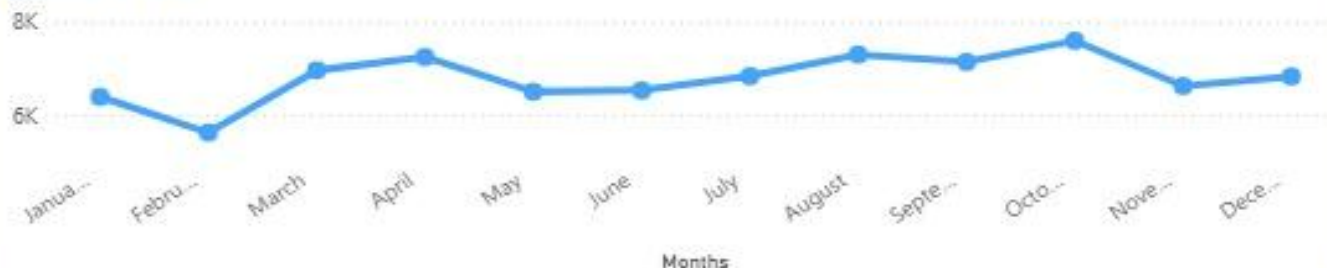
Total Deaths

5K

Road User Type Involved in Deaths or injuries



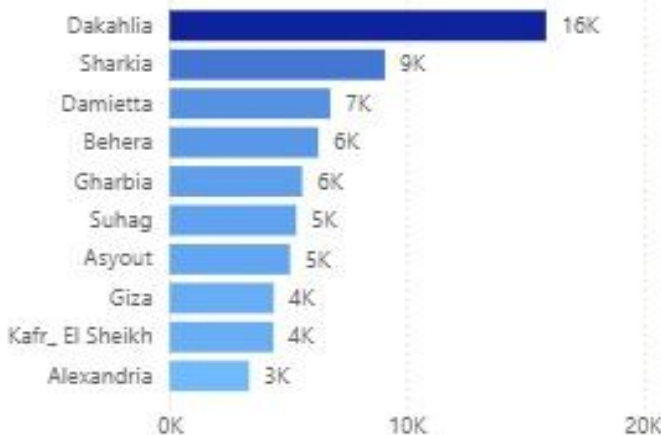
Monthly Trend



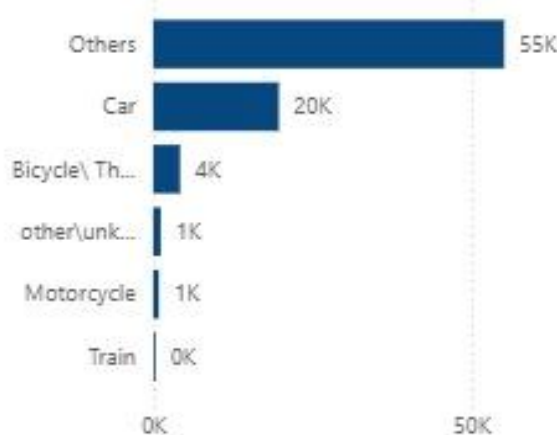
Accident_type by Governorates



Top 10 Governorates



Fatalities by Vehicle Type



Geographical Analysis

Analysis across 27 governorates reveals significant disparities in road safety:

High Risk Zones Sharkia and Dakahlia report the highest casualty and injury volumes.

Rural vs Urban Rural high-speed roads account for severe fatalities despite lower density.

October peaks due to back-to-school traffic, heavier congestion, and changing autumn weather.

Vehicle & Road User Analysis

Road User

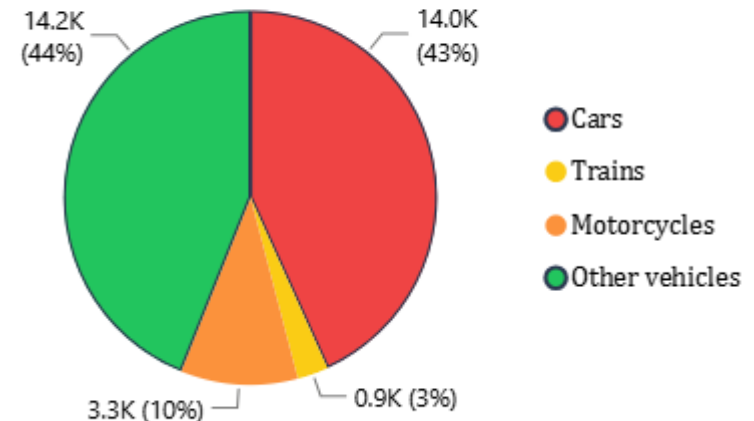
32 LINE ICONS | DISTRACTED & IMPAIRED DRIVING

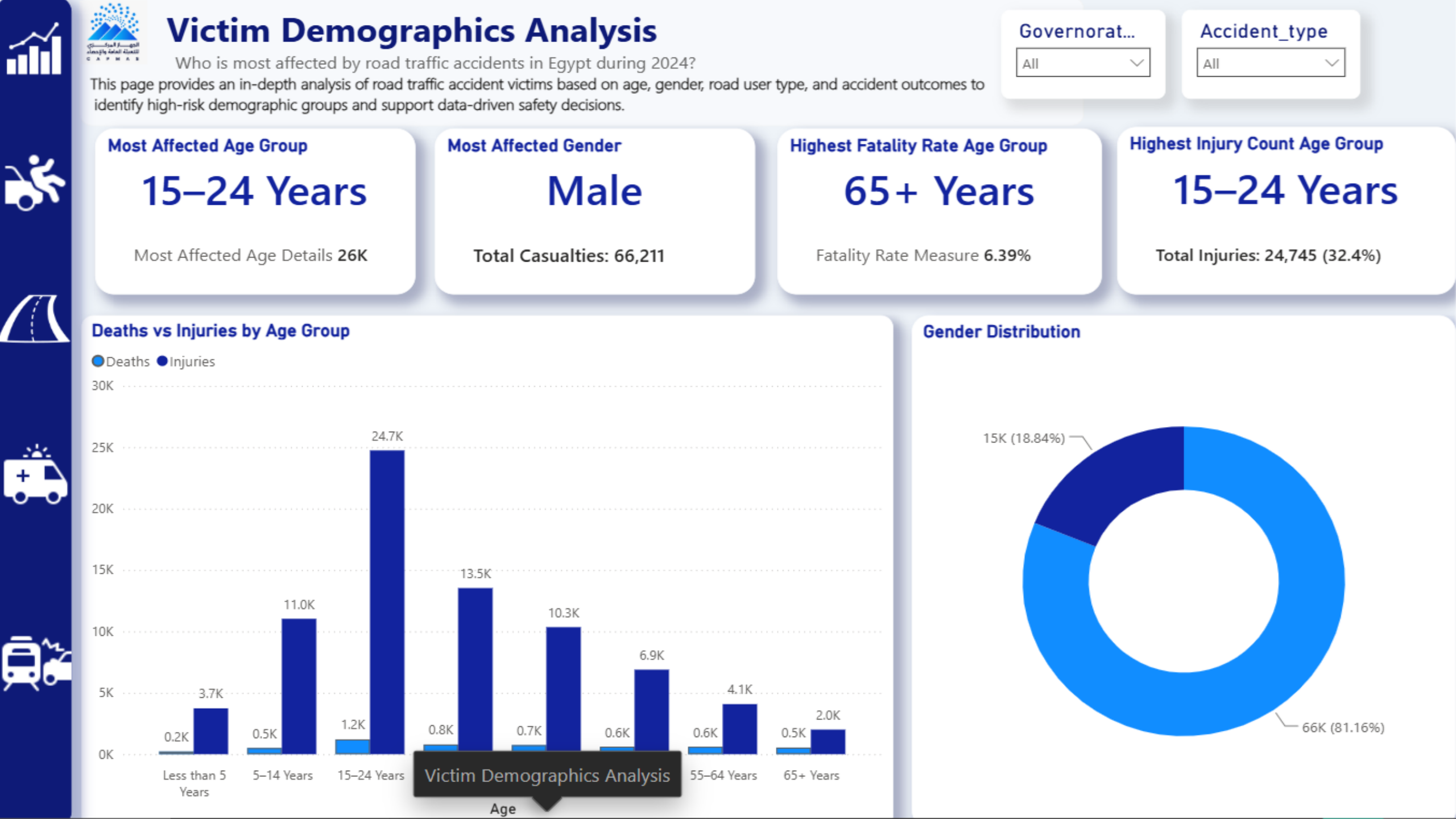
EDITABLE STROKE



Vehicle

Cars, Trains, Motorcycle, and other vehicles





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| Key Insights

10.3%

Fall in fatalities (2024)

15-24

Most affected age group

65+

Highest Fatality group

Insights indicate that while overall deaths are declining, injury rates and train-related incidents are on a rising trend, requiring immediate intervention.

Age Group Analysis

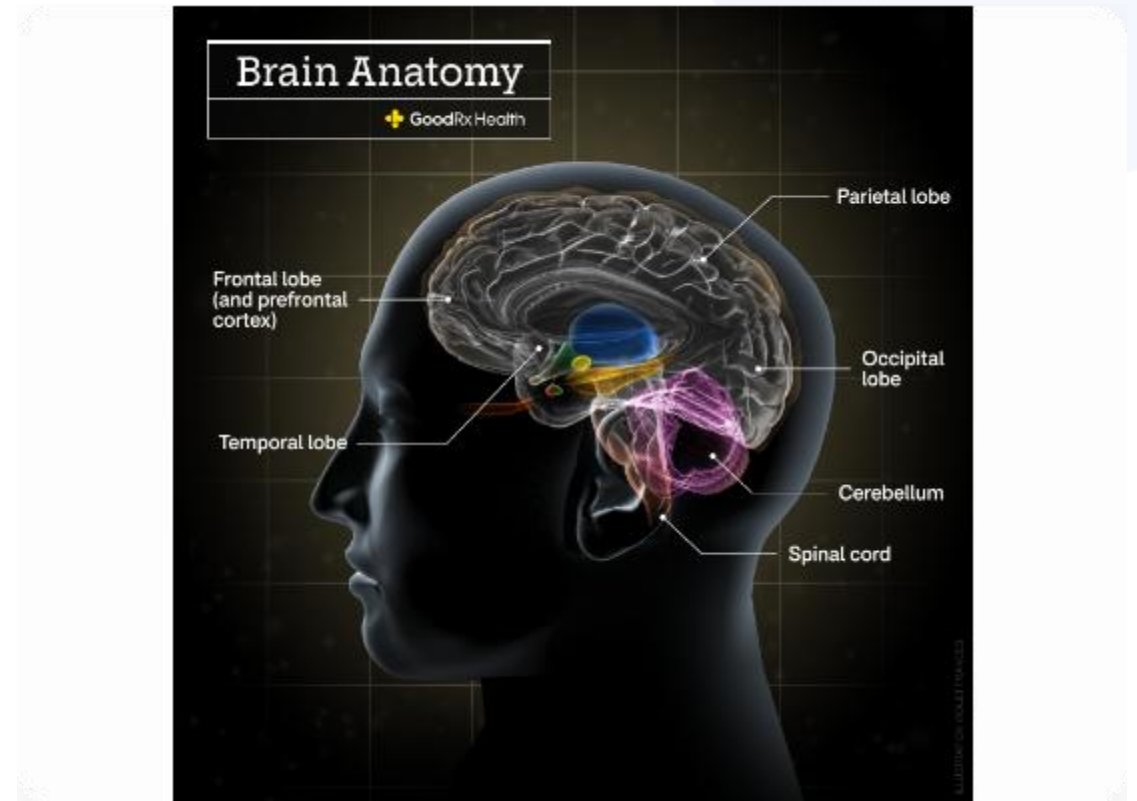
The 15–24 Risk Factor

Neuroscience studies indicate that individuals aged 16 to 25 undergo a developmental stage where the prefrontal cortex continues to mature.

This region handles executive functions like risk assessment and impulse control. An active reward system creates an imbalance, leading to higher rates of speeding and impulsive driving.

The 65+ group:

Due to their physical weakness and sl





High Risk Roads 2024

Which roads and regions have the most identified danger points in Egypt?

This page maps hazard points identified along major roads, based on 422 recorded risk locations across

Governorates

All

Highest Risk Road

القاهرة - الإسكندرية الزراعى

Highest Risk Governorate

Sharkia

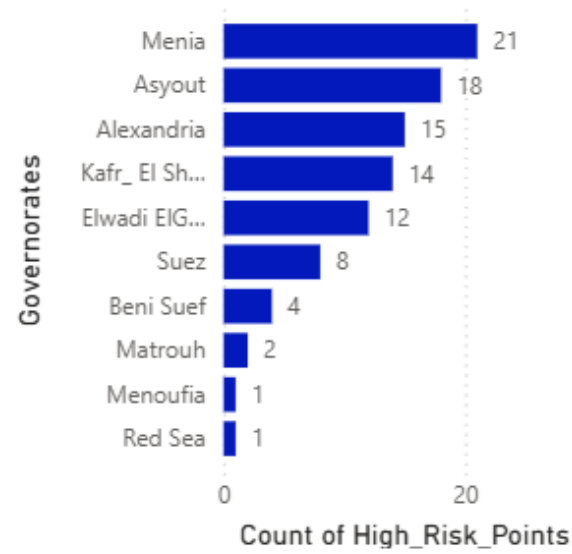
Total Hazard Points

422

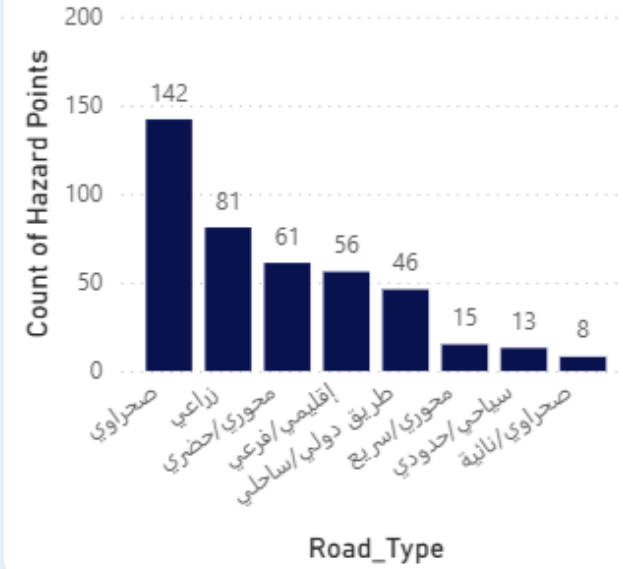
Roads Identified

71

Top 10 Governorates by Hazard Points



Hazard Points by Road Type



Count of Road_ID	Road_Name	Road_Type	Governorates	Total Hazard Points	High_Risk_Points
1	شرم الشيخ الدولي الجديد	طريق دولي/ساحلي	South Sinai	1	كيلو إتجاه النفق 10
1	القاهرة - الإسكندرية الزراعى	زراعي	Behera	1	أبو حمص
1	القاهرة - بليس الصحراوى	صحراوى	Sharkia	1	أبو سمران
1	الدائرى الإقليمى	محوري/حضري	Kalyoubia	1	إتجاه الأسكندرية
1	الطريق الدائرى	محوري/حضري	Cairo	1	إتجاه التجمع
1	كاترين - نوبيع	سياحي/حدودي	South Sinai	1	إتجاه التحرير
1	الدائرى الإقليمى	محوري/حضري	Sharkia	1	إتجاه الصنافين
1	شرم الشيخ الدولي الجديد	طريق دولي/ساحلي	South Sinai	1	إتجاه الطور
1	الدائرى الإقليمى	محوري/حضري	Sharkia	1	إتجاه العاشر
1	القاهرة - الإسماعيلية - بورسعيد الصحراوى	صحراوى	Ismailia	1	إتجاه القاهرة
1	القاهرة - الإسماعيلية - بورسعيد الصحراوى	صحراوى	Ismailia	1	إتجاه القاهرة منطقة السحر والجمال

High Risk Roads

The Most Dangerous Road: The **Cairo - Alexandria Agricultural Road** (الإسكندرية الزراعي - القاهرة) was identified as the highest-risk road with the most prominent danger points.

Top Risk Locations: **Sharkia** is flagged as the highest risk governorate overall, while **Menia** tops the chart for the highest count of specific hazard points with **21 points**.

Hazard Points by Road Type: **Desert roads** (صحراوي) present the highest infrastructure risk with **142 hazard points**, followed by **agricultural roads** (زراعي) with **81 points**.



Emergency Response & Ambulance Coverage

Is ambulance coverage adequate relative to population across governorates?

This page evaluates ambulance infrastructure (cars, points, and coverage rates by population) across Egyptian governorates in 2024

Governorates

All



Total Ambulance Cars

3K

Total Ambulance Points

2K

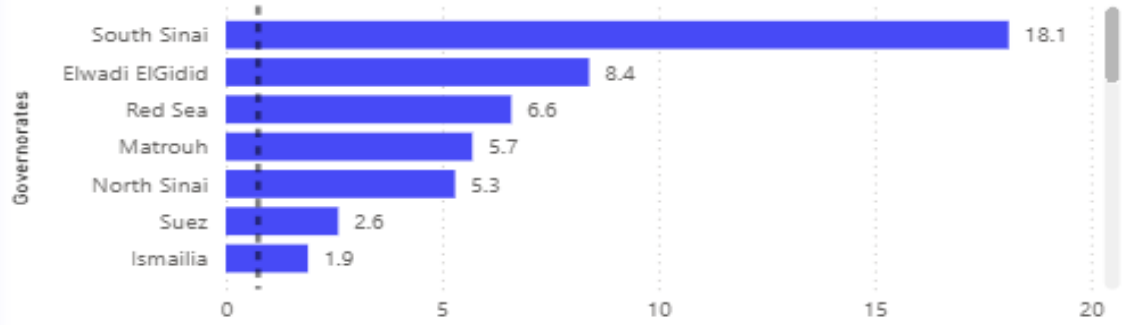
Lowest Coverage Governorate

Menia — 0.4 cars per 25k

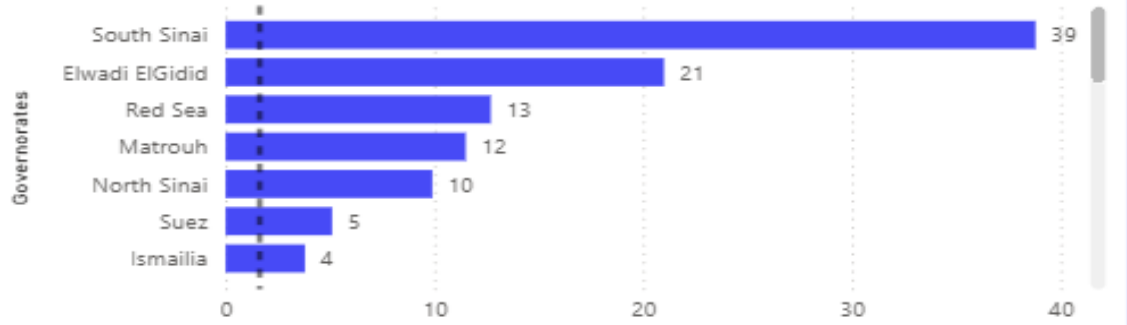
National Average Coverage

0.75

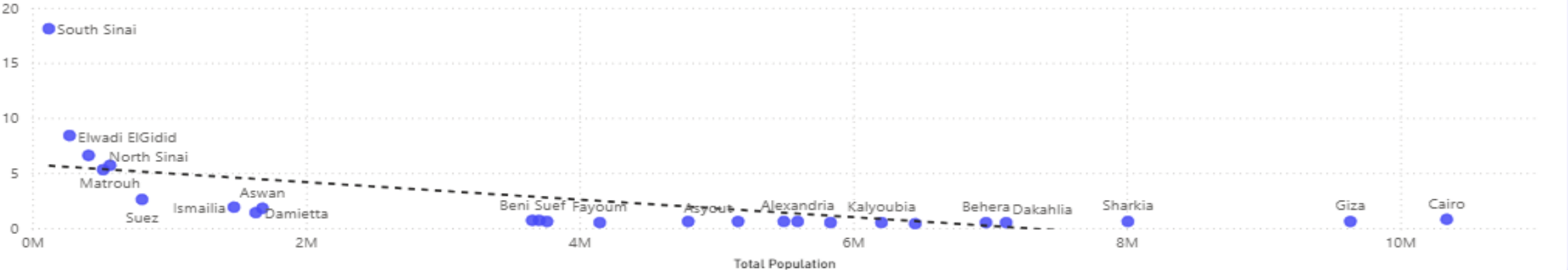
Ambulance Cars per 25,000 People by Governorate



Ambulance Centers per 100,000 People by Governorate



Ambulance Coverage vs Population by Governorate



Emergency & Ambulance Centers and Cars

Inverse Relationship: High-population cities (Cairo/Giza) have the lowest ambulance coverage due to dense population.

Highest Coverage: South Sinai tops Egypt with 18.1 cars and 39 centers per 100K people.

Lowest Coverage: Menia is the most underserved (0.4 cars), correlating with its high road hazard points.



Train Accidents & Safety Analysis 2024

What causes train accidents, and who bears the human cost?

Analyzing formal train accidents alongside person-related incidents, by cause, liability, and region.

Main_Region

All

Accident Categ...

Train Accidents

Deaths

57

Injuries

104

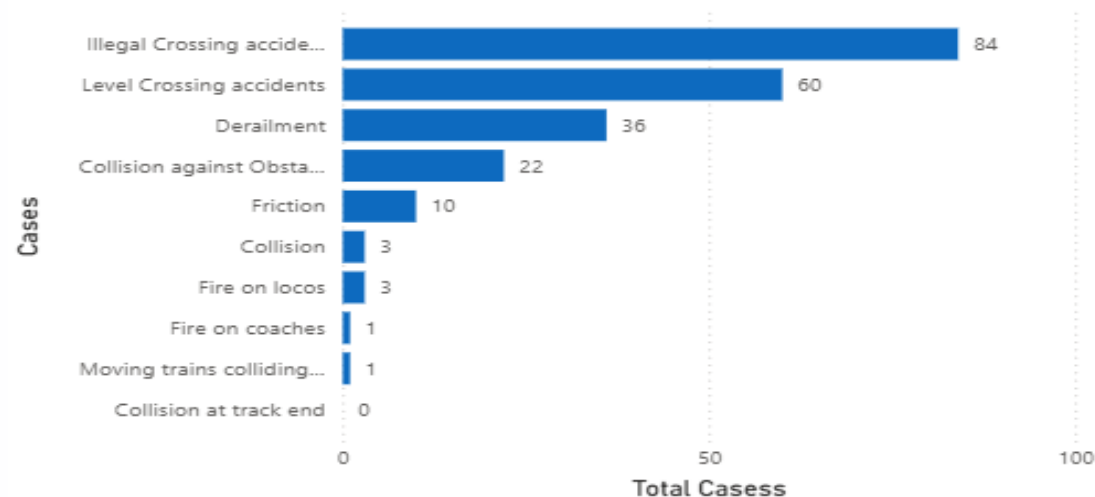
Non Serious Cases

180

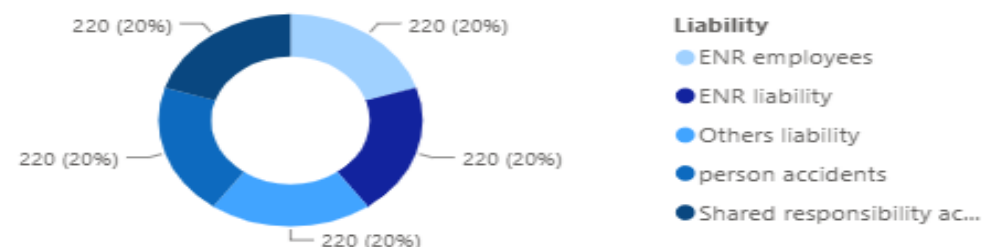
Serious Cases

40

Total Casess by Cases

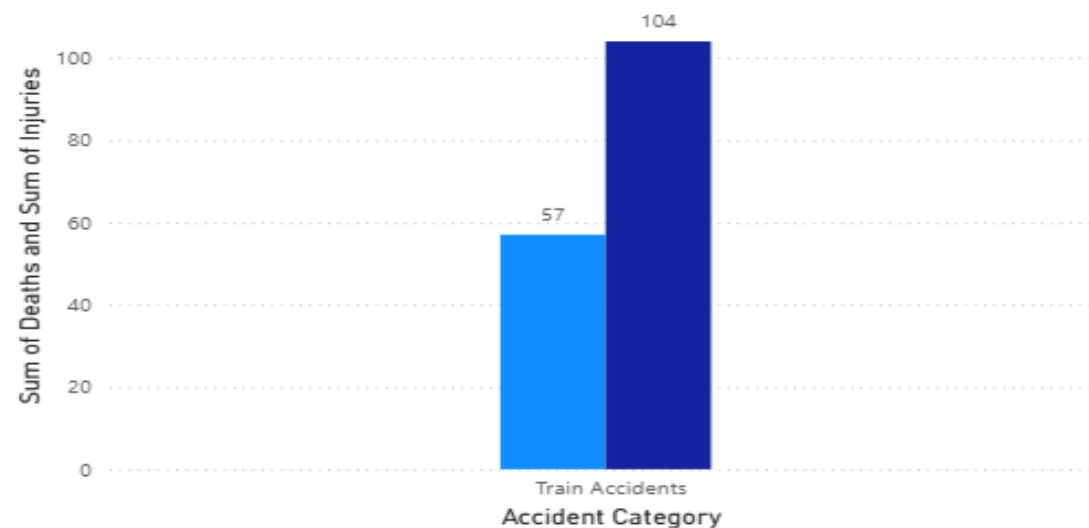


Total Casess by Liability



Sum of Deaths and Sum of Injuries by Accident Category

Sum of Deaths Sum of Injuries



Total Casess by Region



2024 Train Accidents

Train Casualties: 57 deaths, 104 injuries, 180 non-serious, and 40 serious cases.

Primary Cause: Pedestrian negligence leads, with Illegal Crossings (84) and Level Crossings (60) topping cases.

Most Affected Regions: Mid Region ranks highest with 49 cases, followed closely by Cairo (40).

Liability Distribution: Evenly split (20% each) across five parties, showing complex, shared accountability.

Technology Stack



Power BI

Modeling & Dashboards

Power Query

Data ETL & Cleaning

Excel

Validation & Organization

Python

Pandas & Matplotlib

Strategic Recommendations

Monitoring

Speed caps for novice drivers and increased surveillance.



Infrastructure

Regular maintenance and clear traffic signals.



Awareness

Public education on speeding and rules compliance.



Training

Enhanced license testing and safety programs.



Vehicle Safety

Apply strict safety standards and inspections.



Data Usage

Data-driven deployment of emergency resources.

Driving Change Through Data

"Data analysis transforms accident records into actionable insights that can support safer roads and better decision-making."

Thank You